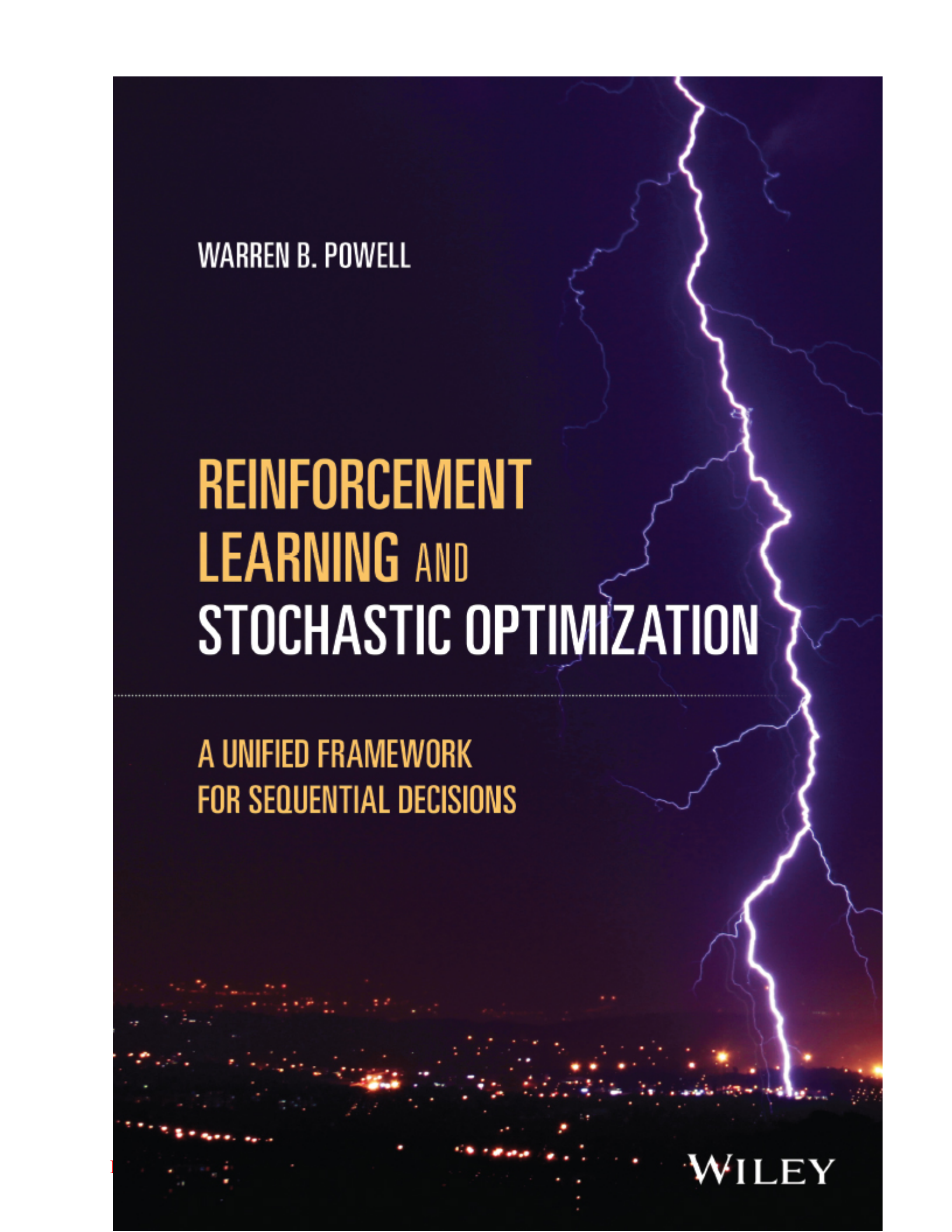




The background of the book cover is a photograph of a night sky with a bright, jagged lightning bolt striking down from the top right. Below the lightning, a city skyline is visible, with numerous small, glowing lights representing buildings and streets.

WARREN B. POWELL

REINFORCEMENT LEARNING AND STOCHASTIC OPTIMIZATION

A UNIFIED FRAMEWORK
FOR SEQUENTIAL DECISIONS

WILEY

Reinforcement Learning and Stochastic Optimization

A Unified Framework for Sequential Decisions

Warren B. Powell
Princeton University
Princeton, NJ

WILEY

This edition first published 2022

© 2022 John Wiley & Sons, Inc.

All rights reserved. No part of this publication may be reproduced, stored in a retrieval system, or transmitted, in any form or by any means, electronic, mechanical, photocopying, recording or otherwise, except as permitted by law. Advice on how to obtain permission to reuse material from this title is available at <http://www.wiley.com/go/permissions>.

The right of Warren B. Powell to be identified as the author of this work has been asserted in accordance with law.

Registered Office

John Wiley & Sons, Inc., 111 River Street, Hoboken, NJ 07030, USA

Editorial Office

111 River Street, Hoboken, NJ 07030, USA

For details of our global editorial offices, customer services, and more information about Wiley products visit us at www.wiley.com.

Wiley also publishes its books in a variety of electronic formats and by print-on-demand. Some content that appears in standard print versions of this book may not be available in other formats.

Limit of Liability/Disclaimer of Warranty

The contents of this work are intended to further general scientific research, understanding, and discussion only and are not intended and should not be relied upon as recommending or promoting scientific method, diagnosis, or treatment by physicians for any particular patient. In view of ongoing research, equipment modifications, changes in governmental regulations, and the constant flow of information relating to the use of medicines, equipment, and devices, the reader is urged to review and evaluate the information provided in the package insert or instructions for each medicine, equipment, or device for, among other things, any changes in the instructions or indication of usage and for added warnings and precautions. While the publisher and authors have used their best efforts in preparing this work, they make no representations or warranties with respect to the accuracy or completeness of the contents of this work and specifically disclaim all warranties, including without limitation any implied warranties of merchantability or fitness for a particular purpose. No warranty may be created or extended by sales representatives, written sales materials or promotional statements for this work. The fact that an organization, website, or product is referred to in this work as a citation and/or potential source of further information does not mean that the publisher and authors endorse the information or services the organization, website, or product may provide or recommendations it may make. This work is sold with the understanding that the publisher is not engaged in rendering professional services. The advice and strategies contained herein may not be suitable for your situation. You should consult with a specialist where appropriate. Further, readers should be aware that websites listed in this work may have changed or disappeared between when this work was written and when it is read. Neither the publisher nor authors shall be liable for any loss of profit or any other commercial damages, including but not limited to special, incidental, consequential, or other damages.

A catalogue record for this book is available from the Library of Congress

Hardback ISBN: 9781119815037; ePub ISBN: 9781119815051; ePDF ISBN: 9781119815044; Obook ISBN: 9781119815068.

Evaluation Only. Created with Aspose.Words. Copyright 2003-2026 Aspose Pty Ltd.

Cover image: © DeniseSmit/Getty Images

Cover design by Wiley

Set in 9.5/12.5pt STIXTwoText by Integra Software Services Pvt. Ltd, Pondicherry, India

Contents

[Cover](#)

[Title page](#)

[Copyright](#)

[Preface](#)

[Acknowledgments](#)

[Part I – Introduction](#)

[1 Sequential Decision Problems](#)

[1.1 The Audience](#)

[1.2 The Communities of Sequential Decision Problems](#)

[1.3 Our Universal Modeling Framework](#)

[1.4 Designing Policies for Sequential Decision Problems](#)

[1.4.1 Policy Search](#)

[1.4.2 Policies Based on Lookahead Approximations](#)

[1.4.3 Mixing and Matching](#)

[1.4.4 Optimality of the Four Classes](#)

[1.4.5 Pulling it All Together](#)

[1.5 Learning](#)

[1.6 Themes](#)

[1.6.1 Blending Learning and Optimization](#)

[1.6.2 Bridging Machine Learning to Sequential Decisions](#)

[1.6.3 From Deterministic to Stochastic Optimization](#)

[1.6.4 From Single to Multiple Agents](#)

[1.7 Our Modeling Approach](#)

[1.8 How to Read this Book](#)

[1.8.1 Organization of Topics](#)

[1.8.2 How to Read Each Chapter](#)

[1.8.3 Organization of Exercises](#)

[1.9 Bibliographic Notes](#)

[Exercises](#)

[Bibliography](#)

[2 Canonical Problems and Applications](#)

[2.1 Canonical Problems](#)

[2.1.1 Stochastic Search – Derivative-based and Derivative-free](#)

[2.1.1.1 Derivative-based Stochastic Search](#)

Evaluation Only. Created with Aspose.Words. Copyright 2003-2026 Aspose Pty Ltd.

[2.1.1.2 Derivative-free Stochastic Search](#)
[2.1.2 Decision Trees](#)
[2.1.3 Markov Decision Processes](#)
[2.1.4 Optimal Control](#)
[2.1.5 Approximate Dynamic Programming](#)
[2.1.6 Reinforcement Learning](#)
[2.1.7 Optimal Stopping](#)
[2.1.8 Stochastic Programming](#)
[2.1.9 The Multiarmed Bandit Problem](#)
[2.1.10 Simulation Optimization](#)
[2.1.11 Active Learning](#)
[2.1.12 Chance-constrained Programming](#)
[2.1.13 Model Predictive Control](#)
[2.1.14 Robust Optimization](#)
[2.2 A Universal Modeling Framework for Sequential Decision Problems](#)
[2.2.1 Our Universal Model for Sequential Decision Problems](#)
[2.2.2 A Compact Modeling Presentation](#)
[2.2.3 MDP/RL vs. Optimal Control Modeling Frameworks](#)
[2.3 Applications](#)
[2.3.1 The Newsvendor Problems](#)
[2.3.1.1 Basic Newsvendor – Final Reward](#)
[2.3.1.2 Basic Newsvendor – Cumulative Reward](#)
[2.3.1.3 Contextual Newsvendor](#)
[2.3.1.4 Multidimensional Newsvendor Problems](#)
[2.3.2 Inventory/Storage Problems](#)
[2.3.2.1 Inventory Without Lags](#)
[2.3.2.2 Inventory Planning with Forecasts](#)
[2.3.2.3 Lagged Decisions](#)
[2.3.3 Shortest Path Problems](#)
[2.3.3.1 A Deterministic Shortest Path Problem](#)
[2.3.3.2 A Stochastic Shortest Path Problem](#)
[2.3.3.3 A Dynamic Shortest Path Problem](#)
[2.3.3.4 A Robust Shortest Path Problem](#)
[2.3.4 Some Fleet Management Problems](#)
[2.3.4.1 The Nomadic Trucker](#)
[2.3.4.2 From One Driver to a Fleet](#)
[2.3.5 Pricing](#)
[2.3.6 Medical Decision Making](#)
[2.3.7 Scientific Exploration](#)

[2.3.8 Machine Learning vs. Sequential Decision Problems](#)

[2.4 Bibliographic Notes](#)

[Exercises](#)

[Bibliography](#)

[3 Online Learning](#)

[3.1 Machine Learning for Sequential Decisions](#)

[3.1.1 Observations and Data in Stochastic Optimization](#)

[3.1.2 Indexing Input \$x^n\$ and Response \$y_n^{+1}\$](#)

[3.1.3 Functions We are Learning](#)

[3.1.4 Sequential Learning: From Very Little Data to ... More Data](#)

[3.1.5 Approximation Strategies](#)

[3.1.6 From Data Analytics to Decision Analytics](#)

[3.1.7 Batch vs. Online Learning](#)

[3.2 Adaptive Learning Using Exponential Smoothing](#)

[3.3 Lookup Tables with Frequentist Updating](#)

[3.4 Lookup Tables with Bayesian Updating](#)

[3.4.1 The Updating Equations for Independent Beliefs](#)

[3.4.2 Updating for Correlated Beliefs](#)

[3.4.3 Gaussian Process Regression](#)

[3.5 Computing Bias and Variance*](#)

[3.6 Lookup Tables and Aggregation*](#)

[3.6.1 Hierarchical Aggregation](#)

[3.6.2 Estimates of Different Levels of Aggregation](#)

[3.6.3 Combining Multiple Levels of Aggregation](#)

[3.7 Linear Parametric Models](#)

[3.7.1 Linear Regression Review](#)

[3.7.2 Sparse Additive Models and Lasso](#)

[3.8 Recursive Least Squares for Linear Models](#)

[3.8.1 Recursive Least Squares for Stationary Data](#)

[3.8.2 Recursive Least Squares for Nonstationary Data*](#)

[3.8.3 Recursive Estimation Using Multiple Observations*](#)

[3.9 Nonlinear Parametric Models](#)

[3.9.1 Maximum Likelihood Estimation](#)

[3.9.2 Sampled Belief Models](#)

[3.9.3 Neural Networks – Parametric*](#)

[3.9.4 Limitations of Neural Networks](#)

[3.10 Nonparametric Models*](#)

[3.10.1 K-Nearest Neighbor](#)

[3.10.2 Kernel Regression](#)

[3.10.3 Local Polynomial Regression](#)
[3.10.4 Deep Neural Networks](#)
[3.10.5 Support Vector Machines](#)
[3.10.6 Indexed Functions, Tree Structures, and Clustering](#)
[3.10.7 Comments on Nonparametric Models](#)
[3.11 Nonstationary Learning*](#)
[3.11.1 Nonstationary Learning I – Martingale Truth](#)
[3.11.2 Nonstationary Learning II – Transient Truth](#)
[3.11.3 Learning Processes](#)
[3.12 The Curse of Dimensionality](#)
[3.13 Designing Approximation Architectures in Adaptive Learning](#)
[3.14 Why Does It Work?**](#)
[3.14.1 Derivation of the Recursive Estimation Equations](#)
[3.14.2 The Sherman-Morrison Updating Formula](#)
[3.14.3 Correlations in Hierarchical Estimation](#)
[3.14.4 Proof of Proposition 3.14.1](#)
[3.15 Bibliographic Notes](#)

[Exercises](#)

[Bibliography](#)

[4 Introduction to Stochastic Search](#)

[4.1 Illustrations of the Basic Stochastic Optimization Problem](#)

[4.2 Deterministic Methods](#)

[4.2.1 A “Stochastic” Shortest Path Problem](#)

[4.2.2 A Newsvendor Problem with Known Distribution](#)

[4.2.3 Chance-Constrained Optimization](#)

[4.2.4 Optimal Control](#)

[4.2.5 Discrete Markov Decision Processes](#)

[4.2.6 Remarks](#)

[4.3 Sampled Models](#)

[4.3.1 Formulating a Sampled Model](#)

[4.3.1.1 A Sampled Stochastic Linear Program](#)

[4.3.1.2 Sampled Chance-Constrained Models](#)

[4.3.1.3 Sampled Parametric Models](#)

[4.3.2 Convergence](#)

[4.3.3 Creating a Sampled Model](#)

[4.3.4 Decomposition Strategies*](#)

[4.4 Adaptive Learning Algorithms](#)

[4.4.1 Modeling Adaptive Learning Problems](#)

[4.4.2 Online vs. Offline Applications](#)

[4.4.2.1 Machine Learning](#)

[4.4.2.2 Optimization](#)

[4.4.3 Objective Functions for Learning](#)

[4.4.4 Designing Policies](#)

[4.5 Closing Remarks](#)

[4.6 Bibliographic Notes](#)

[Exercises](#)

[Bibliography](#)

Part II – Stochastic Search

5 Derivative-Based Stochastic Search

[5.1 Some Sample Applications](#)

[5.2 Modeling Uncertainty](#)

[5.2.1 Training Uncertainty \$W^1, \dots, W^N\$](#)

[5.2.2 Model Uncertainty \$S^0\$](#)

[5.2.3 Testing Uncertainty](#)

[5.2.4 Policy Evaluation](#)

[5.2.5 Closing Notes](#)

[5.3 Stochastic Gradient Methods](#)

[5.3.1 A Stochastic Gradient Algorithm](#)

[5.3.2 Introduction to Stepsizes](#)

[5.3.3 Evaluating a Stochastic Gradient Algorithm](#)

[5.3.4 A Note on Notation](#)

This document was truncated here because it was created in the Evaluation Mode.